

Winding Ambiguity in ω Bridging

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Context

Test case: IS-901. $\omega_{\text{true}} = (0.5, -0.3, 2.0)^\circ/\text{s}$, $|\omega_{\text{true}}| = 2.083^\circ/\text{s}$.

Experiments 1–4 use a 120 s window (100 observations, $\Delta t_{\text{samp}} \approx 1.21 \text{ s}$, $\sigma = 0.05 \text{ mag}$). Experiments 5–6 use a 3600 s window (500 observations) with brightness peaks at indices 183, 260, 360.

Experiments

Experiment 1: Forward Propagation — Analytic ω , Exact Attitude, Hi-fi

Variables	None (no free parameters)
Constraints	IS-901; q_{k-1}, q_k exact from true trajectory; initial orientation: 160° about arbitrary axis; hi-fi shadowing
Residual	MSE between reconstructed and observed LC (computed, not optimised)
Solver	None — $\omega = \text{rotvec}(q_{k-1}^{-1} q_k) / \Delta t_{\text{samp}}$ (analytic, two consecutive observations); Euler forward propagation from q_k

Result: Reconstructed LC overlays truth. At $\Delta t_{\text{samp}} \approx 1.21 \text{ s}$ the axis-angle estimate recovers ω to $< 10^{-6}^\circ/\text{s}$ error because the rotation between consecutive observations ($\sim 2.5^\circ$) has no winding ambiguity.

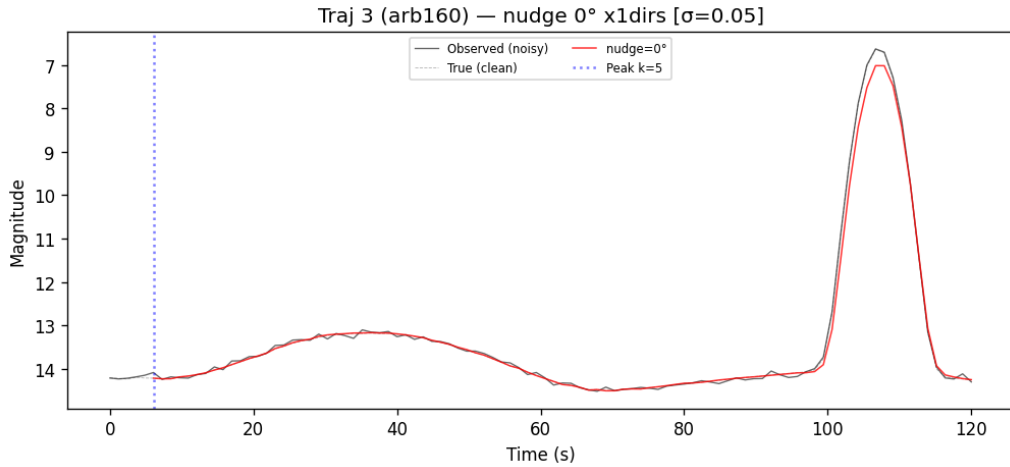


Figure 1: Hi-fi forward propagation from exact q_k with analytic ω (trajectory 3, arb160). Red overlays truth.

Experiment 2: Forward Propagation — Analytic ω , 3° Attitude Nudge, Hi-fi

Variables	Nudge direction (5 random unit vectors, deterministic seed 123; not optimised)
Constraints	Same analytic ω as Exp 1 (from exact q); q_k nudged 3° before propagation; hi-fi shadowing
Residual	MSE per direction (computed, not optimised)
Solver	None — same analytic ω ; Euler forward propagation from nudged q_k

Result: Sensitivity is anisotropic. Near the glint ($t \approx 107$ s), some nudge directions barely change the LC; others shift the peak by >2 mag.

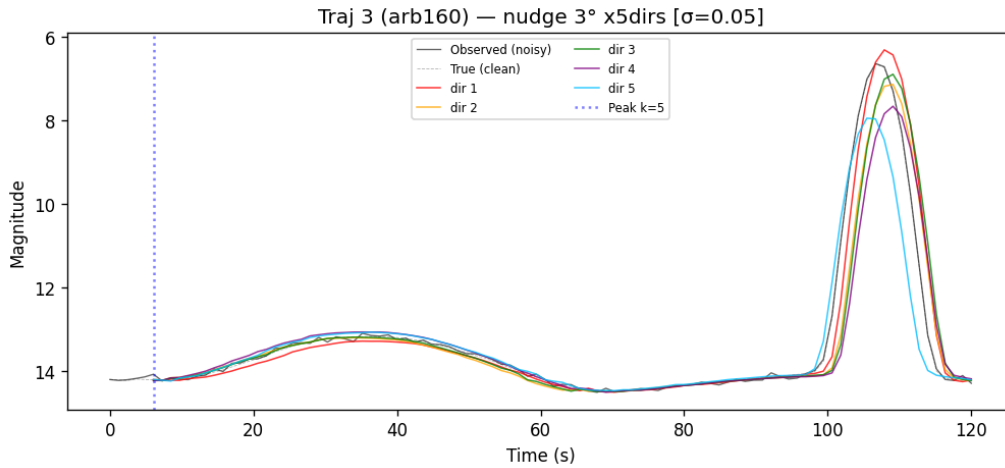


Figure 2: Hi-fi forward propagation from 3°-nudged q_k in 5 directions (trajectory 3, arb160).

Experiment 3: Forward Propagation — Analytic ω , Exact Attitude, Lo-fi Reconstruction

Variables	None
Constraints	Same analytic ω and exact q_k as Exp 1; reconstruction uses lo-fi (no self-shadowing); observed LC is hi-fi; orientation: arb160
Residual	MSE(lo-fi reconstruction, hi-fi observed)
Solver	None — forward evaluation only

Result: Lo-fi predicts a bright glint at $t \approx 60$ s that does not exist in the hi-fi observations. At this orientation a facet is in self-shadow; hi-fi suppresses it, lo-fi does not.

Experiment 4: Forward Propagation — Analytic ω , 3° Nudge, Lo-fi (Different Orientation)

Variables	Nudge direction (5 random unit vectors)
Constraints	Same analytic ω ; q_k nudged 3°; lo-fi reconstruction; orientation: 30° about $\hat{x}$
Residual	MSE(lo-fi reconstruction, hi-fi observed) per direction
Solver	None — forward evaluation only

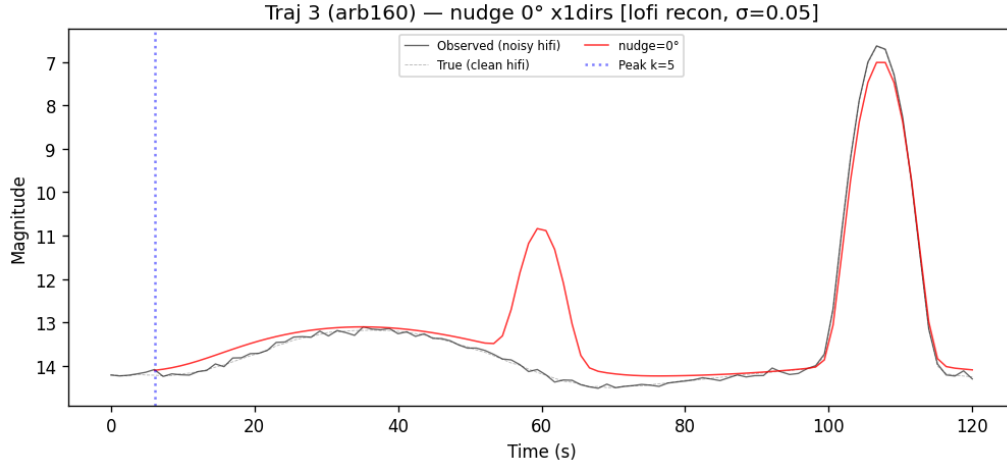


Figure 3: Lo-fi reconstruction (no shadows) vs hi-fi observed (trajectory 3, arb160). Phantom peak at $t \approx 60$ s.

Result: No phantom peak. Lo-fi and hi-fi agree at this orientation. The lo-fi failure in Exp 3 occurs only at orientations with active self-shadowing and cannot be predicted a priori.

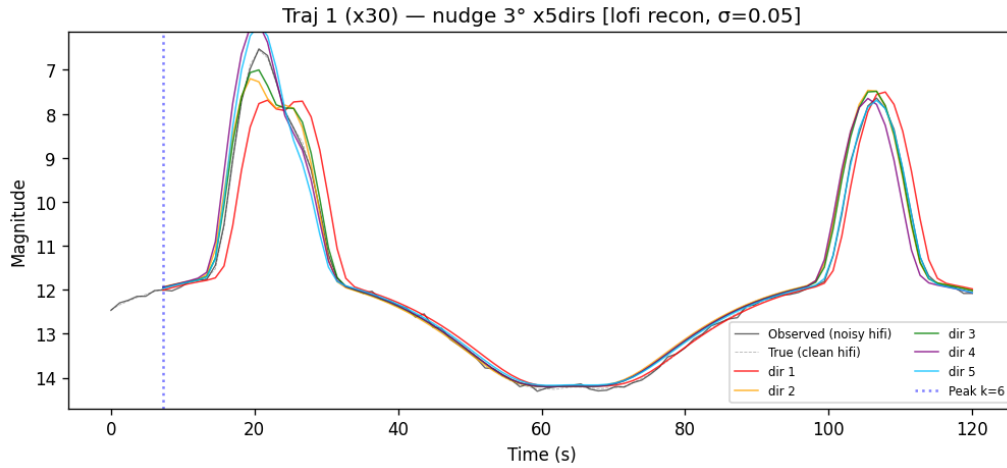


Figure 4: Lo-fi reconstruction at benign orientation (trajectory 1, x30). No phantom peak.

Experiment 5: Staircase ω Enumeration via Barrier-Constrained Optimisation

Variables	$\omega \in \mathbb{R}^3$ (optimised per step)
Constraints	q_A, q_B exact (oracle) at peaks 183, 260; $\Delta t = 555.5$ s; soft lower barrier on $ \omega ^2$ incremented after each step to force progression through winding bands
Residual	Optimisation objective: $1 - (q_{\text{prop}} \cdot q_B)^2 + \lambda_{\text{barrier}}$; post-hoc trough score: $ b_{\text{pred}} - b_{\text{ref}} $ (lo-fi, single epoch between peaks)
Solver	L-BFGS-B, 8 sequential steps; step 0 init: axis-angle $\text{rotvec}(q_A^{-1} q_B) / \Delta t$; subsequent steps init: previous ω + one extra revolution along rotation axis

Result:

Step	$ \omega $ ($^\circ/\text{s}$)	Arrival err	Trough err
0	0.316	0.0	1.719
1	0.959	0.0	0.099
2	1.601	0.0	0.021 $\leftarrow$ best trough
3	2.230	0.0	0.792 $\leftarrow$ true ($ \omega_{\text{true}} = 2.083$)
4	2.883	0.036	0.044
5	3.509	0.0	0.134
6	4.180	0.0	1.749
7	4.824	0.0	1.491

Staircase successfully enumerates 8 distinct winding solutions. Trough scoring (lo-fi, single epoch) selects step 2 (wrong)—at the trough epoch, steps 2 and 3 have not diverged enough to produce a detectable brightness difference.

Experiment 6: Angular Momentum Conservation Filter — Two-Leg Staircase

Variables	$\omega \in \mathbb{R}^3$ per leg per step (optimised independently)
Constraints	q_A, q_{mid}, q_B exact (oracle) at peaks 183, 260, 360; leg 0 $\Delta t = 555.5$ s; leg 1 $\Delta t = 721.4$ s; lower + upper $ \omega $ barriers per step ($\pm 0.8 \delta$ band)
Residual	Per step: $1 - (q_{\text{prop}} \cdot q_{\text{end}})^2 + \text{barrier penalties}$; L-filter: $\ \mathbf{L}_{\text{leg}0}[k] - \mathbf{L}_{\text{leg}1}[j]\ $ where $\mathbf{L} = R(q_{\text{mid}}) I \omega$
Solver	L-BFGS-B per step (with barriers); exhaustive $8 \times 8 = 64$ L-comparison at shared peak 260

Result: Filter selected ($k=0, j=0$). True pair: ($k=3, j=1$). **Incorrect**—leg 1 staircase is broken.

Step	Leg 0 $ \omega $ ($^\circ/\text{s}$)	Leg 1 $ \omega $ ($^\circ/\text{s}$)
0	0.316	0.249
1	0.959	3.279 $\leftarrow$ gap
2	1.601	3.509
3	2.230	3.995

Leg 0 steps uniformly ($\sim +0.65^\circ/\text{s}$ per step). Leg 1 jumps from 0.249 to 3.279, skipping winding bands 1–3. Cause: axis-angle init for leg 1 lands near a π -rotation; quaternion double-cover causes discontinuous jump. Fix: initialise from $\omega_0 = \mathbf{0}$, tighten upper barrier.

Decisions

- MIP/integer programming ruled out (consulted Jack—insufficient discrete structure).
- Lo-fi cannot be used for scoring near glints (Exp 3–4). Hi-fi required.
- Single-trough scoring insufficient (Exp 5). Need multi-epoch or physics-based filter.
- Angular momentum conservation is the most promising ω filter (Exp 6), pending staircase fix.

Next Steps

1. Fix staircase seed ($\omega_0 = \mathbf{0}$ + tight upper barrier) for leg 1.
2. Rerun L-conservation filter with fixed staircase.
3. Test with approximate endpoints (2–5° attitude error).
4. Stack multiple troughs as secondary discriminator.